Locating all Lagrange points with a curvature-learning MemComputing machine

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The MemComputing paradigm solves a problem by mapping its solutions onto the stable fixed points of a dynamical system that is augmented with memory degrees of freedom. Applied mostly to combinatorial optimization, it transfers naturally to a classic continuous problem: the five Lagrange points of the restricted three-body problem. A damped gradient flow on the Jacobi effective potential settles onto its minima—the triangular points L_4, L_5—but is repelled by the collinear saddles L_1, L_2, L_3, whose transverse direction is unstable. We turn the saddles into attractors by inverting the force along that direction, a single sign that must be $+1$ in the saddle's unstable direction and -1 at the stable minima. We show that *no constant choice works*, so this sign is the load-bearing ingredient—and we let a memory supply it. The memory accumulates a quasi-Newton (Broyden) estimate of the local curvature from the sequence of gradient evaluations the machine already performs, with *no* analytic second derivative, and from it sets the inversion. The resulting autonomous flow with memory locates all five Lagrange points of every Sun–planet and planet–moon pair of the solar system, twenty-three systems with mass ratios from $\mu \approx 2 \times 10^{-9}$ to 0.1. The construction is an instructive, fully reproducible example in which a memory degree of freedom performs a genuine computation—learning the curvature of a landscape—that no static rule can replace.

I. INTRODUCTION AND SCOPE

MemComputing is a non-von-Neumann paradigm in which a physical system exploits *time non-locality*—the ability of internal degrees of freedom to retain and use information about their past—to solve a problem[1]. In a Digital MemComputing Machine the solutions of the target problem are mapped onto the stable fixed points of a continuous dynamical system, and trajectories launched from generic initial conditions are guided to a solution in an augmented phase space whose coordinates are the relaxed original variables *plus* the memory degrees of freedom[2,3]. The paradigm has been applied mostly to combinatorial optimization[4]; here we use it for a classic continuous problem of celestial mechanics, and in doing so isolate a clean example of a memory variable doing genuine computational work.

The restricted three-body problem (RTBP)—two primaries on a circular orbit and a massless test particle—and its five Lagrange equilibria are a cornerstone of nonlinear dynamics and of mission design[5,6]. The equilibria are the critical points of the Jacobi effective potential Ω. A dissipative gradient flow is the natural dynamical-system route to such points, but it has a basic limitation: damping drives a flow only to *minima* of Ω. Two of the five Lagrange points (the triangular L_4, L_5) are minima and are found this way; the other three (the collinear L_1, L_2, L_3) are *saddles*, and a damped flow is pushed away from them along their unstable direction. Making a saddle into an attractor requires inverting the force along that direction—turning the unstable direction into a restoring one. This single sign is what the machine must supply, and supplying it correctly is the whole problem: the sign depends on the local curvature of Ω, which is positive at the minima and negative along the saddles' unstable direction.

Our central observation is that *no constant sign works*: inverting always breaks the minima, never inverting never captures the saddles. The sign must track the local curvature, and we let a memory degree of freedom do so. Crucially, the memory does not read the analytic curvature; it *learns* it, accumulating a quasi-Newton estimate of the Hessian from the very gradient evaluations the flow already performs, exactly as a Broyden or BFGS iteration builds curvature information from a history of gradients[8,9]. The resulting machine—a single autonomous flow whose force inversion is controlled by a curvature-learning memory—locates all five Lagrange points of every two-body pair in the solar system, twenty-three systems spanning eight orders of magnitude in mass ratio, with no analytic second derivative anywhere in the dynamics. We list every parameter, give the off-axis controls that prove the memory is load-bearing, and discuss honestly what the approach does and does not buy.

II. THE RESTRICTED THREE-BODY PROBLEM

We use the co-rotating frame in normalized units: the two primaries are separated by unit distance, the orbital angular velocity of the frame is $\omega = 1$, and the mass ratio is

$$\mu = \frac{M_2}{M_1 + M_2} \in (0, \tfrac{1}{2}], \tag{1}$$

placing M_1 at $(-\mu, 0)$ and M_2 at $(1-\mu, 0)$. We define the effective potential

$$\Omega(x, y) = \frac{x^2 + y^2}{2} + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \tag{2}$$

with $r_1 = \sqrt{(x+\mu)^2 + y^2}$ and $r_2 = \sqrt{(x-1+\mu)^2 + y^2}$. This is the negative of the Jacobi potential commonly written with overall minus signs; with the convention (2) the Lagrange points are the critical points of Ω, and—as we use below—the triangular points are its *minima*. The gradient is

$$\partial_x \Omega = x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3}, \tag{3}$$

$$\partial_y \Omega = y - \frac{(1-\mu)\,y}{r_1^3} - \frac{\mu\,y}{r_2^3}. \tag{4}$$

A Lagrange point is a relative equilibrium, so the condition reduces to

$$\boxed{\nabla\Omega(x, y) = \mathbf{0}} \tag{5}$$

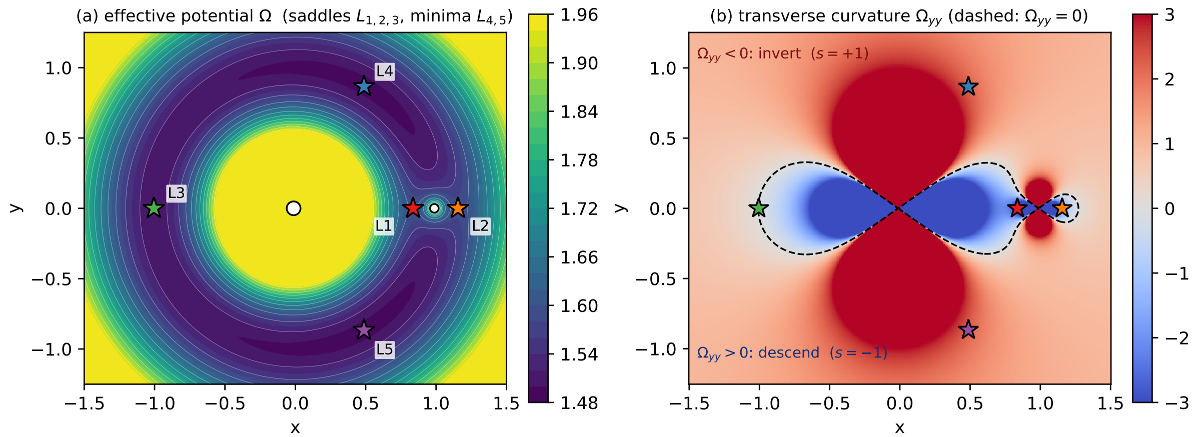

FIG. 1. The effective potential and its curvature for the Earth–Moon system ($\mu = 0.0121$). **(a)** $\Omega(x, y)$, Eq. (2); the collinear L_1, L_2, L_3 are saddles and the triangular L_4, L_5 are minima; the large and small white dots are the primary and secondary. **(b)** the transverse curvature Ω_{yy}, Eq. (6) (clipped at ± 3), with the contour $\Omega_{yy} = 0$ dashed. Where $\Omega_{yy} < 0$ (red) the machine inverts the transverse force ($s = +1$) to make the saddle attract; where $\Omega_{yy} > 0$ (blue) it descends normally ($s = -1$). The machine never reads this map; it learns the sign from its own gradient history.

with exactly five solutions: the collinear L_1, L_2, L_3 on the axis $y = 0$ and the triangular $L_{4,5} = (\frac{1}{2} - \mu, \pm\frac{\sqrt{3}}{2})^5$.

The *nature* of each critical point—which is what the machine must contend with—is set by the Hessian of Ω. Its transverse component is

$$\Omega_{yy} = 1 - \frac{1-\mu}{r_1^3} + \frac{3(1-\mu)y^2}{r_1^5} - \frac{\mu}{r_2^3} + \frac{3\mu y^2}{r_2^5}. \tag{6}$$

On the axis ($y = 0$) this reduces to $1 - (1-\mu)/r_1^3 - \mu/r_2^3$, which is *negative* at all three collinear points; together with $\Omega_{xx} = 1 + 2(1-\mu)/r_1^3 + 2\mu/r_2^3 > 0$ and $\Omega_{xy} = 0$ there, the collinear points are *saddles*, unstable in the transverse (y) direction. At the triangular points one finds $\Omega_{xx} = \frac{3}{4}$, $\Omega_{yy} = \frac{9}{4}$, and $\Omega_{xy} = \frac{3\sqrt{3}}{4}(1-2\mu)$, giving $\det \mathrm{Hess}\, \Omega = \frac{27}{4}\mu(1-\mu) > 0$ with positive trace: L_4, L_5 are *minima* of Ω. The sign of Ω_{yy} thus distinguishes the two cases: it is negative in the unstable direction of the collinear saddles and positive at the triangular minima (Fig. 1). The machine's task is to read this sign correctly—ultimately, to learn it—using only information generated along its own trajectory.

III. FINDING EQUILIBRIA BY LANDSCAPE INVERSION

Consider a damped flow that descends Ω in the rotating frame,

$$\ddot{x} = 2\dot{y} - \partial_x \Omega - \gamma \dot{x}, \tag{7a}$$

$$\ddot{y} = -2\dot{x} + s\, \partial_y \Omega - \gamma \dot{y}, \tag{7b}$$

with constant viscosity $\gamma > 0$ and an *inversion gain* s multiplying the transverse force. With $s = -1$ the transverse force is the ordinary descent term $-\partial_y \Omega$, which is restoring at a minimum; the flow then settles onto the minima of Ω, i.e. the triangular points L_4, L_5. At a collinear saddle, however, the transverse direction has $\Omega_{yy} < 0$: a small displacement $y > 0$ gives $\partial_y \Omega < 0$, so $-\partial_y \Omega > 0$ pushes the particle *away* from the axis and the saddle repels. Setting $s = +1$ reverses this single term into a restoring force, converting the saddle's unstable direction into a stable one—the saddle becomes a minimum of the effective, inverted landscape and an attractor of the flow.

The correct rule is therefore $s = -\operatorname{sign}(\Omega_{yy})$: invert ($s = +1$) where the transverse curvature is negative (the saddles), descend normally ($s = -1$) where it is positive (the minima). The key point is that this sign is *intrinsically position-dependent*, and *no constant value of s can work*. We verify this directly. Seeding the flow off the symmetry axis—so that the transverse direction is actually excited—around a collinear saddle (L_1) and around a triangular minimum (L_4) of the Earth–Moon system, we record how many of twelve seeds reach the target under each rule (Table I). A constant $s = -1$ reaches the minimum from all seeds but *never* the saddle; a constant $s = +1$ reaches the saddle but *breaks* the minimum, reaching it from no seed. Only the curvature-adaptive sign captures both. The force inversion is thus genuinely load-bearing—unlike the viscosity γ, which merely removes energy and for which, we note, any constant value in a broad range suffices.

TABLE I. No constant inversion works. Off-axis seeds (twelve each) around the Earth–Moon collinear saddle L_1 and triangular minimum L_4; entries are the number reaching the target under the raw dynamics (7) (no refinement). “adaptive” is $s = -\operatorname{sign}(\Omega_{yy})$; “memory” is the Hessian-free machine of Sec. IV.

seed region	$s \equiv -1$	$s \equiv +1$	adaptive	memory
L_1 (collinear saddle)	0/12	12/12	10/12	11/12
L_4 (triangular minimum)	12/12	0/12	12/12	12/12

IV. A MEMORY THAT LEARNS THE CURVATURE

The adaptive rule of Sec. III requires the sign of Ω_{yy}. Reading it from the analytic Hessian (6) works, but it is unsatisfying: the machine would be handed the very curvature information that distinguishes the solutions. We instead let a memory degree of freedom *learn* it. The flow already evaluates the gradient $\mathbf{g} = \nabla\Omega$ at every step, and the Jacobian of $\mathbf{g}$ is the Hessian. A memory matrix B therefore accumulates a *quasi-Newton* estimate of the Hessian from successive gradient samples. With $\Delta\mathbf{p} = \mathbf{p}_k - \mathbf{p}_{k-1}$ and $\Delta\mathbf{g} = \mathbf{g}_k - \mathbf{g}_{k-1}$, the step from B_k to B_{k+1} adds a Broyden secant assimilation[8,9] and a slow relaxation toward the identity,

$$B_{k+1} = B_k + \frac{(\Delta\mathbf{g} - B_k\,\Delta\mathbf{p})\,\Delta\mathbf{p}^{\mathsf{T}}}{\Delta\mathbf{p}^{\mathsf{T}}\Delta\mathbf{p}} + \lambda\,(\mathbb{I} - B_k)\,\Delta t, \tag{8}$$

applied only where $\|\Delta\mathbf{p}\|$ exceeds a small gate (otherwise $B_{k+1} = B_k$); the right-hand side is then symmetrized to respect the symmetry of a Hessian. The two added terms are of different character. The first is the secant correction, which enforces $B\,\Delta\mathbf{p} = \Delta\mathbf{g}$ along the current step: a full per-step assimilation whose size is set by how wrong B is and is *independent* of Δt (Broyden is a discrete iteration, with no continuum limit as Δt shrinks to zero). The second is a genuine rate—a slow leak toward the identity. Because the curvature varies across the landscape, this leak gives B a finite memory horizon $\sim 1/\lambda$, so it tracks the *local* curvature rather than a path-history average, and relaxing toward $\mathbb{I}$ supplies a safe default ($B_{yy} = 1$) where the step is too small to measure. Equivalently, B follows a continuous leak $\dot{B} = -\lambda(B - \mathbb{I})$ punctuated by discrete secant measurements—the structure of a continuous–discrete (Kalman-type) estimator. The estimated transverse curvature is the component B_{yy}, and the inversion gain is a second memory variable $s \in [-1, 1]$ that relaxes toward the learned sign,

$$\dot{s} = \alpha\Big[-\tanh\Big(\tfrac{B_{yy}-\theta_s}{\varepsilon_c}\Big) - s\Big]. \tag{9}$$

The augmented phase space is thus $(x, y, \dot{x}, \dot{y},\ s,\ B)$: the relaxed mechanical variables plus the memory (s, B). The full machine integrates (7) with γ constant, advancing B by (8) and s by (9) at each step. No analytic second derivative appears anywhere in the dynamics; Ω_{yy} is used only, after the fact, to confirm that the learned B_{yy} is correct.

Two features make the scheme work. First, it is *predictive*. During the off-axis approach to a saddle the trajectory samples gradients over a range of positions, so B accumulates a usable curvature estimate *before* the trajectory reaches the unstable region—it does not have to wait for the runaway to develop. (A purely reactive detector, one that infers instability from the growth of the motion, is outrun by the exponential instability of the saddle and fails; the quasi-Newton accumulation is what makes the memory fast enough.) Second, the threshold $\theta_s < 0$ in (9) biases s toward the safe descending state $s = -1$, inverting only when the learned curvature is *clearly* negative. This protects the minima and the small-μ corotation ridge (where $\|\nabla\Omega\| \approx 0$ along $r = 1$ and the curvature estimate hovers near zero) while still inverting at genuine saddles, where B_{yy} is strongly negative. The bias is needed because the damped rotating frame is not symmetric under sign reversal of y—the Coriolis term breaks it—so L_4 and L_5 have slightly different basins; biasing toward the safe state treats both robustly.

Figure 2 shows the memory at work on two Earth–Moon trajectories. On the approach to the collinear saddle L_1 the estimate B_{yy} converges to a negative value tracking the true Ω_{yy}, and s ramps from -1 to $+1$, inverting the force and capturing the saddle. On the approach to the triangular minimum L_4 the estimate is positive, s stays at -1, and the flow descends normally. The same memory, with no curvature supplied, discovers opposite signs in the two cases.

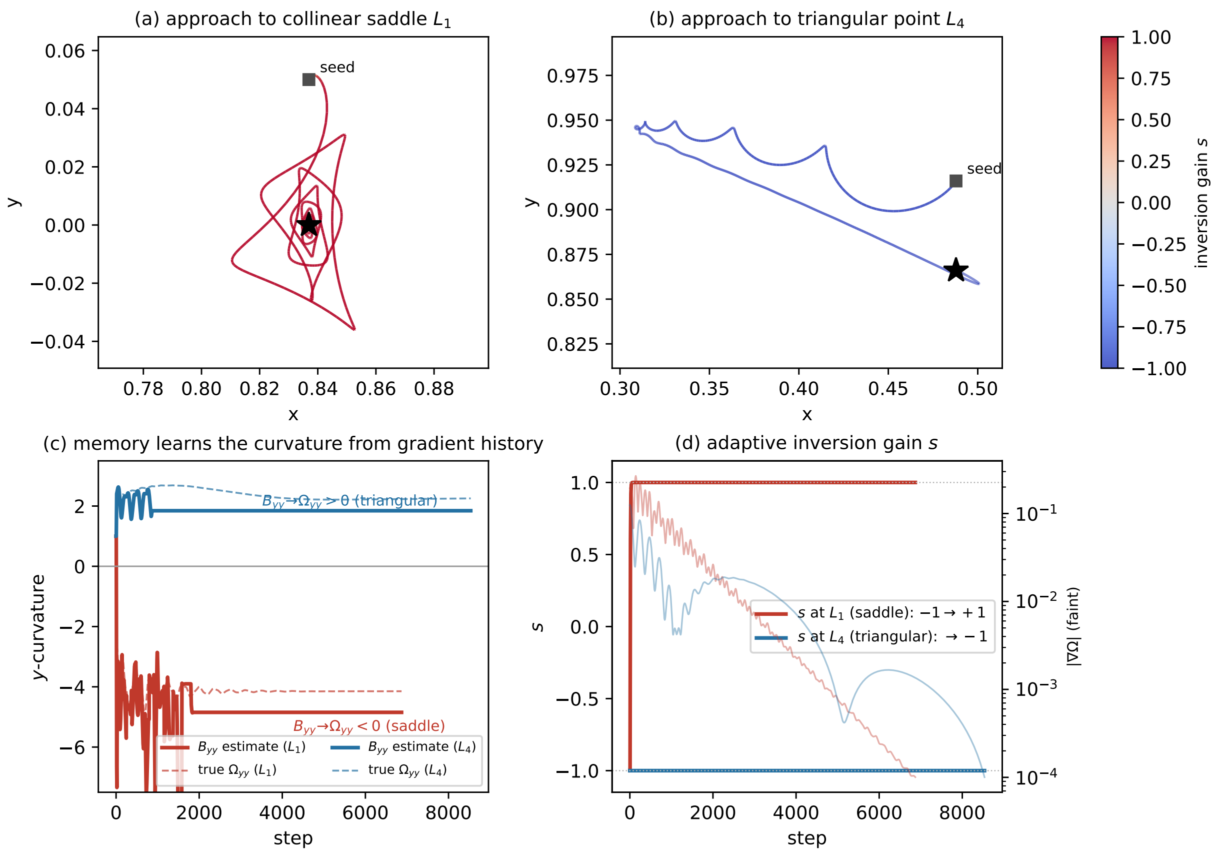

FIG. 2. The memory learns the curvature. Two Earth–Moon trajectories of the same Hessian-free machine. **(a),(b)** phase-plane paths coloured by the inversion gain s: red ($s = +1$) on the approach to the collinear saddle L_1, blue ($s = -1$) on the approach to the triangular minimum L_4. **(c)** the quasi-Newton estimate B_{yy} (solid) tracks the true Ω_{yy} (dashed) from gradient history alone—negative at the saddle, positive at the minimum. **(d)** the gain s responds to what was learned: from -1 to $+1$ at the saddle, held at -1 at the minimum, while $\|\nabla\Omega\|$ (faint) decays to convergence.

V. NUMERICAL IMPLEMENTATION AND PARAMETERS

Everything below is the procedure actually integrated in the accompanying code[10]; the parameters are collected in Table II.

a. Integration. The equations are advanced by explicit Euler with fixed step Δt; a guard $\varepsilon = 10^{-9}$ is added to r_1, r_2. The memory is initialized at $B = \mathbb{I}$, $s = -1$ (the safe descending state), and the Broyden update (8) is applied only when $\|\Delta\mathbf{p}\|$ exceeds a small gate, so that near a fixed point—where $\Delta\mathbf{p}$ vanishes—the estimate simply freezes. A trajectory stops when $\|\nabla\Omega\| < \theta$, when it leaves the disk $\|\mathbf{r}\| > 12$, or after $N_{\max}$ steps.

b. Seed grid. No solution coordinates are given to the integrator. Starting positions are generated from the *generic structure* of the RTBP only, using the Hill radius $r_H = (\mu/3)^{1/3}$ and the secondary position $x_\mathrm{s} = 1 - \mu$: (A) on-axis seeds $(x_\mathrm{s} \pm c\, r_H, 0)$ for $c \in \{0.5, 0.7, 1, 1.4, 2, 3, 5, 8\}$, bracketing the collinear L_1, L_2; (B) L_3 seeds near $x = -1$; (C) equilateral seeds about $(\frac{1}{2}, \pm\frac{\sqrt{3}}{2})$ for the triangular points; (D) a coarse off-axis fill for the broad large-μ basins. Seeds inside the exclusion disks around either primary are discarded, leaving of order seventy seeds per system. Even the equilateral seeds use the fixed $(\frac{1}{2}, \pm\frac{\sqrt{3}}{2})$ rather than the μ-dependent solution; the dynamics and the refinement below produce the exact positions.

c. Refinement and labelling. Each stopped trajectory is optionally polished by a few Newton iterations on $\nabla\Omega = \mathbf{0}$, accepted only if it reaches $\|\nabla\Omega\| < 10^{-9}$, otherwise the raw endpoint is kept. A converged point is identified with the nearest analytic Lagrange position if within 0.05 (polished) or 0.12 (raw); these analytic positions are used only for post-hoc identification, never by the machine.

TABLE II. Parameters used for all results, identical for every system.

Symbol	Meaning	Value
γ	viscosity (constant)	0.3
α	relaxation rate of the gain s	15
ε_c	curvature scale in Eq. (9)	0.3
λ	Broyden leak toward $\mathbb{I}$	2.0
θ_s	safe-state bias (invert if $B_{yy} < \theta_s$)	-0.6
$\|\Delta\mathbf{p}\|_{\min}$	Broyden update gate	10^{-4}
Δt	Euler time step	0.01
θ	convergence threshold $\|\nabla\Omega\|$	10^{-4}
$N_{\max}$	maximum steps	1.2×10^5
ε	singularity guard on r_1, r_2	10^{-9}
accept	Newton refinement acceptance	$\|\nabla\Omega\| < 10^{-9}$

VI. RESULTS

The central result is that the curvature-learning machine, with the single parameter set of Table II and *no analytic Hessian in its dynamics*, discovers all five Lagrange points of

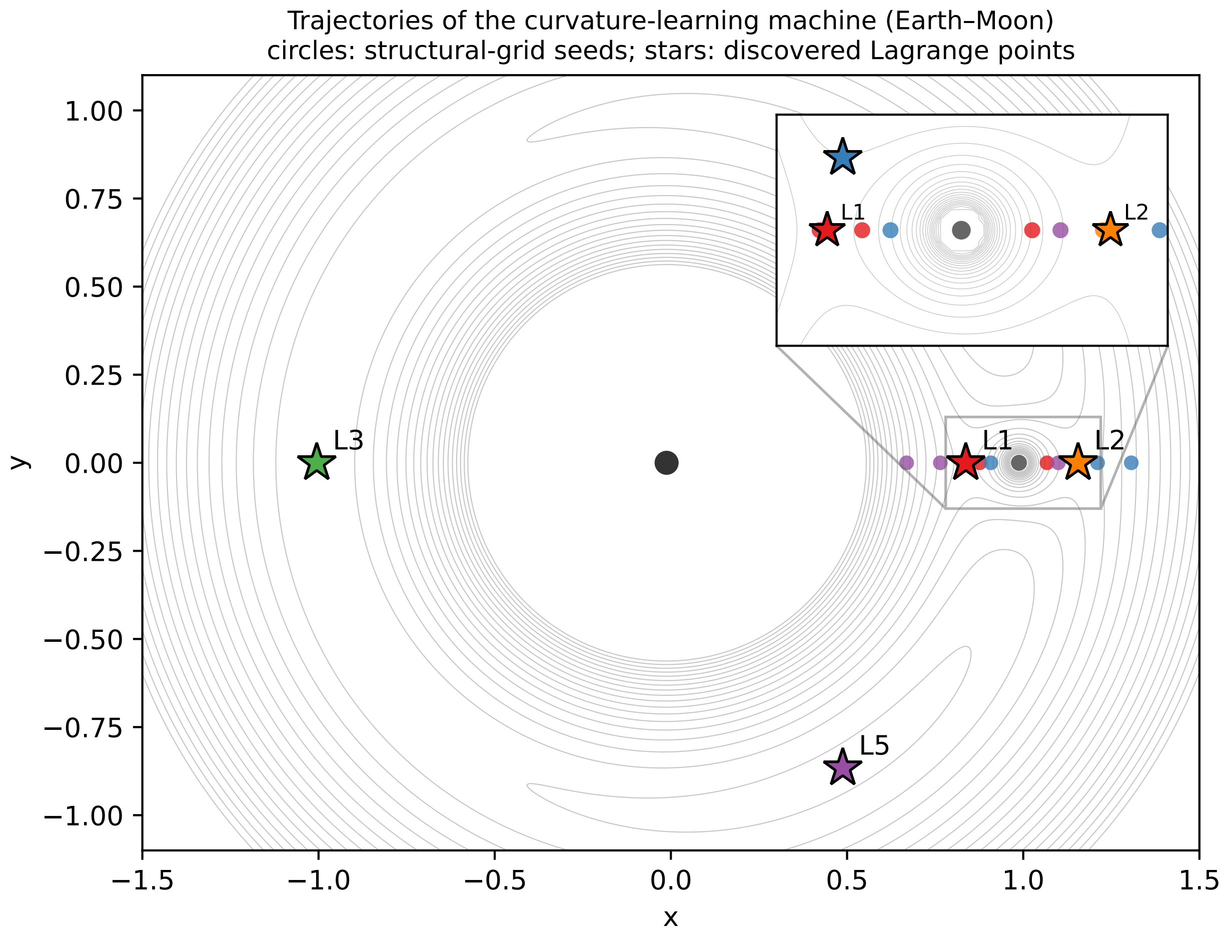

FIG. 3. Trajectories of the curvature-learning machine for the Earth–Moon system, each coloured by the Lagrange point it converges to (stars); circles mark the structural-grid seeds and grey curves are level sets of Ω. The inset zooms the cramped L_1/L_2 region near the secondary. The machine receives no solution coordinates and no analytic curvature.

every two-body system in the solar system. Table III lists the twenty-three Sun–planet and planet–moon pairs, with mass ratios spanning eight orders of magnitude, from Mars–Deimos ($\mu = 2.3 \times 10^{-9}$) to Pluto–Charon ($\mu = 0.11$). Every system returns 5/5; after the optional Newton polish the located positions agree with the analytic Lagrange points to better than 10^{-9}. Figure 3 shows trajectories of the machine reaching all five points of the Earth–Moon system from the structural seed grid.

The off-axis controls of Table I establish the role of the memory: it supplies a sign that no constant can. The smaller-μ systems are the demanding ones—the triangular minima sit on the nearly flat corotation ridge—and they are exactly where the safe-state bias θ_s matters; with it, the Hessian-free machine matches the analytic-curvature machine across the whole range. For reference, the identical machine driven by the analytic Ω_{yy} instead of the learned B_{yy} also returns 5/5 on all twenty-three systems, so the gradient-history estimate loses nothing.

TABLE III. All five Lagrange points found for every solar-system pair by the curvature-learning machine (no analytic Hessian in the dynamics), using the single parameter set of Table II. "Stable" refers to the linear stability of L_4, L_5 (Routh, $\mu < 0.03852$[7]).

System	μ	Found	L_4/L_5 stable
Mars–Deimos	2.3×10^{-9}	5/5	yes
Sun–Pluto	6.6×10^{-9}	5/5	yes
Mars–Phobos	1.7×10^{-8}	5/5	yes
Sun–Mercury	1.7×10^{-7}	5/5	yes
Saturn–Enceladus	1.9×10^{-7}	5/5	yes
Sun–Mars	3.2×10^{-7}	5/5	yes
Sun–Venus	2.4×10^{-6}	5/5	yes
Sun–Earth	3.0×10^{-6}	5/5	yes
Saturn–Rhea	4.1×10^{-6}	5/5	yes
Jupiter–Europa	2.5×10^{-5}	5/5	yes
Uranus–Oberon	3.5×10^{-5}	5/5	yes
Uranus–Titania	4.1×10^{-5}	5/5	yes
Sun–Uranus	4.4×10^{-5}	5/5	yes
Jupiter–Io	4.7×10^{-5}	5/5	yes
Sun–Neptune	5.1×10^{-5}	5/5	yes
Jupiter–Callisto	5.7×10^{-5}	5/5	yes
Jupiter–Ganymede	7.8×10^{-5}	5/5	yes
Neptune–Triton	2.1×10^{-4}	5/5	yes
Saturn–Titan	2.4×10^{-4}	5/5	yes
Sun–Saturn	2.9×10^{-4}	5/5	yes
Sun–Jupiter	9.5×10^{-4}	5/5	yes
Earth–Moon	1.2×10^{-2}	5/5	yes
Pluto–Charon	1.1×10^{-1}	5/5	no

VII. WHY A MEMORY, AND WHAT IT DOES NOT DO

It is worth being precise about where the memory is, and is not, essential. The viscosity γ in (7) removes energy and quenches the motion, but this is a generic role: any constant

γ in a broad range damps the flow onto the attractors equally well, and a time-varying or memory-controlled viscosity buys nothing here. The inversion gain s is different. As Table I shows, the sign it carries cannot be constant—it must be $+1$ in the saddles' unstable direction and -1 at the minima—so a degree of freedom that adapts it to the local landscape is genuinely required. The memory is that degree of freedom, and in its quasi-Newton form it does not merely store a number it was given: it *computes* the curvature, accumulating it from the history of gradient evaluations the flow already performs. This is the sense in which the memory is the active ingredient of the computation, and it is the clean lesson the example teaches.

The construction has clear limits, which we state plainly. It does not compete with classical root finders on speed: a Newton iteration converges in a handful of evaluations once a bracket is known, far faster than integrating a trajectory, and we retain an optional Newton polish for the final digits. The advantage is of a different kind. The machine is *global and prior-knowledge-free*: one autonomous flow, one parameter set, locates all five equilibria of any system from a generic grid, without a bracket per point and without being told which points are saddles. More broadly, it finds critical points of *any index*, not only minima—the force inversion turns a saddle into an attractor—which ordinary gradient descent and the optimization methods built on it cannot do. Here the inversion is a single transverse sign because the collinear saddles' unstable direction is the y-axis; in general the same memory matrix B supplies the unstable eigendirections, and the force would be inverted along them. The Lagrange points are a transparent, curriculum-central testbed for that general primitive, with the bonus that the memory's job—learning a curvature it is never told—can be watched directly (Fig. 2).

VIII. CONCLUSION

We have introduced a MemComputing machine that locates all five Lagrange points of the restricted three-body problem by descending the Jacobi effective potential and inverting the force along the unstable direction of its saddles. The inversion sign cannot be constant—no fixed choice captures both the saddles and the minima—so it is carried by a memory degree of freedom, which *learns* the local curvature as a quasi-Newton estimate accumulated from the machine's own gradient evaluations, with no analytic second derivative in the dynamics. The machine discovers all five points of every Sun–planet and planet–moon pair in the solar system—twenty-three systems from $\mu \approx 2 \times 10^{-9}$ to 0.1—from a generic structural grid and a single fixed parameter set. The example extends the MemComputing paradigm into celestial mechanics and provides a transparent, fully reproducible case in which a memory performs a genuine computation, learning the curvature of a landscape, that no static rule can replace. The code and an interactive implementation are available[10].

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